

Pseudocode

① Design a class hierarchy

START

CLASS Person

Name, Age

END CLASS

CLASS Student INHERITS Person

RegNo, Maths[5]

METHOD Average ()

$Avg = (M_1 + M_2 + M_3 + M_4 + M_5) / 5$

END METHOD

END CLASS

CLASS ResearchStudent INHERITS Student

Research Area

METHOD display ()

PRINT Name, Age, RegNo,

Research Area, Avg

END METHOD

END CLASS

CREATE ResearchStudent

INPUT details

CALL Average ()

CALL display ()

STOP

2) Complete the Missing Pseudocode:

```
BEGIN  
  READ N  
  READ ARRAY A  
  FOR i ← 0 TO N-1  
    FOR j ← i+1 TO N-1  
      IF A[i] == A[j]  
        DELETE A[j]  
      END IF  
    END FOR  
  END FOR  
  PRINT Modified Array  
END.
```

3) Debug the Pseudocode

```
BEGIN  
  CLASS Area  
    METHOD area(side)  
      RETURN side * side  
    END METHOD  
    METHOD area(length, breadth)  
      RETURN length * breadth  
    END METHOD  
  END CLASS  
END.
```

④ Trace the Execution

BEGIN

sum $\leftarrow$ 0

For $i \leftarrow 1$ TO 5

IF $i \bmod 2 == 0$

sum $\leftarrow$ sum + i

ELSE

sum $\leftarrow$ sum - i

ENDIF

ENOFOR

PRINT sum

END.

⑤ Write Efficient Pseudocode:

BEGIN

READ N

READ Array A

first $\leftarrow -\infty$

second $\leftarrow -\infty$

For $i \leftarrow 0$ TO N-1

IF $A[i] > \text{first}$ THEN

second $\leftarrow$ first

first $\leftarrow A[i]$

ELSE IF $A[i] > \text{second}$ AND $A[i] \neq$

first THEN

second $\leftarrow A[i]$

ENDIF

ENOFOR

PRINT second

END.

⑥ Convert Problem statement into Pseudocode

BEGIN

INTERFACE Payment

METHOD Pay(amount)

END INTERFACE

CLASS CreditCard IMPLEMENTS Payment

Pay(amount)

END CLASS

CLASS UPI IMPLEMENTS Payment

Pay(Amount)

END CLASS

CLASS NetBanking IMPLEMENTS Payment

Pay(amount)

END CLASS

READ choice, amount

IF choice = 1

Payment ← CreditCard

ELSE IF choice = 2

Payment ← UPI

ELSE

Payment ← NetBanking

ENDIF

CALL Payment.Pay(amount)

END.

7) complete the Pseudocode

```
BEGIN
CLASS Animal
METHOD speak()
PRINT "Animal"
END METHOD
END CLASS
CLASS Dog INHERITS Animal
METHOD speak()
PRINT "Barks"
END METHOD
END CLASS
Animal obj ← NEW Dog()
obj.speak()
END.
```

8) Find Logical ERRORS

```
BEGIN
CLASS Employee
DATA
    id
    name
CONSTRUCTOR(id, name)
    this.id ← id
    this.name ← name
END CONSTRUCTOR
END CLASS
END.
```

Case Study (Integrated Problem)

BEGIN

CLASS Vehicle

VehicleNo, ownerName

METHOD display()

END CLASS

CLASS Car INHERITS Vehicle

Fuel Type

END CLASS

CLASS Truck INHERITS Vehicle

loadCapacity

END CLASS

READ N

FOR $i \leftarrow 1$ TO N

READ type

IF type = "Car"

obj $\leftarrow$ NewCar()

ELSE

obj $\leftarrow$ NewTruck()

END IF

READ VehicleNo, ownerName

STORE obj

ENDFOR

FOR EACH obj

CALL obj.display()

ENDFOR

END